BREEZE: A Training-Free Closed-Loop Physical-Gate Admission Framework for LLM-Generated Bearing Fault Signals

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ABSTRACT

Large language models (LLMs) can synthesize condition-monitoring signals without training a dedicated generative model, but prompt-level physical priors remain open-loop. A generated window may enter the augmentation pool even when its statistics, spectral energy, or envelope-spectrum structure are not supported by the real training data for the requested fault class. This paper proposes BREEZE, a training-free closed-loop admission framework that wraps a black-box generator with deterministic signal-processing gates calibrated only on the real training split. The active gates check format legality, robust time-statistic support, train-supported spectral shape, envelope-spectrum fault evidence, reliability-gated current-sideband evidence, and diversity. Failed generations are converted into structured reports for resampling; no waveform repair or generator training is performed. On the Paderborn University benchmark under the N09_M07_F10 operating condition, the original K=3 closed loop accepts 78.89% of 450 generation slots with 2.05 LLM calls per slot. A v2 verifier is evaluated as an offline rescreening of the same 450 slots: it admits 286 slots, produces 761 accepted items before diversity, keeps 757 windows after diversity screening, and provides a more conservative gate-compliance record. Compared with real-only training, the v2 pool improves test accuracy by 14.33, 7.12, and 5.70 percentage points for 10, 25, and 50 real windows per class, respectively; these three comparisons remain significant after Benjamini–Hochberg correction over the full family of 42 v2-vs-main-baseline paired tests. The v2 pool is competitive with, but does not uniformly outperform, strong conventional augmentation and open-loop physics-guided baselines. Experiments on a private four-channel machine-tool dataset are reported as a schema-level verifier portability check, not as an augmentation claim, because machine speed/geometry metadata and an audited synthetic pool are unavailable. The results support train-calibrated physical-gate admission as a practical and auditable way to control black-box synthetic data, while also defining its empirical limits.

1. Introduction

Fault data scarcity remains a bottleneck for intelligent condition monitoring. Industrial machines rarely run to failure, controlled seeded-fault campaigns are expensive, and fault-class imbalance is severe. Synthetic data therefore play an important role in few-shot fault diagnosis, with methods ranging from noise and geometric augmentation to variational autoencoders, generative adversarial networks, and diffusion models [1, 2, 3].

LLM-based time-series synthesis offers a different route. A general-purpose language model can be prompted to produce signal recipes or numeric sequences conditioned on a machine state, avoiding the cost of training a new generator for each dataset. The preceding MechaForge study showed that LLM-produced signals can improve downstream fault diagnosis in low-data regimes, but it also exposed a critical limitation: the generation process is open-loop. A prompt may describe bearing physics, yet the emitted signal can still lack class-consistent envelope-spectrum evidence, realistic impulsiveness, or supported vibration-current structure before it enters the training pool.

Most existing physical approaches solve this problem at training time, by embedding physics losses into a GAN, VAE, or diffusion objective. That is powerful but not plug-and-play: a new machine, sensor configuration, or closed-source generator requires re-training or re-design. Simple post-generation filters are training-free but often stop at moment matching, which misses diagnostic structure such as BPFO/BPFI alignment and resonance band evidence. What is missing is an inference-time wrapper that can test a candidate against train-calibrated diagnostic evidence, produce actionable feedback, and admit only samples that pass the active gates, without modifying the generator or repairing the waveform after the fact.

This paper proposes BREEZE, a training-free closed-loop physical verification framework. BREEZE controls which generated samples are admitted into the synthetic training pool. Its verifier is built from signal-processing and bearing-diagnostic operations: robust train-split boundaries, Welch spectral features, spectral-kurtosis-supported demodulation bands, Hilbert envelope-spectrum checks, current sideband scores, and nearest-neighbour diversity admission. All thresholds are calibrated from the real training split only; the test set is never used for parameter selection. Failed samples produce structured constraint reports that can be fed back to the generator for at most K resampling rounds. Samples that still fail are rejected rather than locally corrected.

The word "certificate" is used in this manuscript only in the engineering sense of an auditable gate-compliance record. It is not a formal proof of physical correctness, distributional coverage, or downstream generalization.

The contributions are:

1. A training-free, generator-agnostic physical-gate admission framework that wraps an existing LLM signal generator at inference time, using no generator retraining and no post-hoc waveform repair.
2. A bearing-physics verifier with train-supported statistical, spectral, envelope, current-sideband, and diversity checks. Each admitted sample receives an auditable per-sample gate report.
3. A v2 offline verifier design that reduces high-risk hard-band decisions using smooth spectral features, PSD-CDF Wasserstein distance, multi-band resonance support, and robust union domains. v2 is evaluated by rescreening the frozen $K=3$ generation pool, not by running a new v2 LLM loop.
4. A reproducible evaluation protocol separating physical fidelity, admission cost, and downstream diagnostic utility on the Paderborn benchmark, plus schema-level verifier checks on additional PU operating conditions and a private four-channel machine-tool dataset.

2. Related Work

2.1. Synthetic data for fault diagnosis

Classical augmentation methods inject noise, rescale amplitudes, crop windows, or mix samples. They are cheap and often strong baselines, as confirmed in this study, but they cannot create new fault mechanisms. Trainable generative models address this limitation by learning the real signal distribution: GANs [1], VAEs [2], and diffusion models [3] have all been adapted to time-series augmentation. Sequence generators such as C-RNN-GAN [4], recurrent conditional GANs [5], and TimeGAN [6] explicitly model temporal dynamics, while diffusion models such as TimeGrad [7] and CSDI [8] extend denoising or score-based generation to multivariate time series.

For machinery diagnosis, generative augmentation is attractive because seeded fault data are expensive and imbalanced. Conditional GAN/VAE/diffusion pipelines can synthesize vibration windows for minority classes and have been reported to improve classifiers in low-data regimes. These methods are the closest family of competitors to BREEZE, but they optimize the generator itself. Therefore, their fidelity depends on architecture, training data, loss weighting, and convergence. BREEZE targets a different operating point: it does not train or fine-tune the generator, and it asks whether generated windows should be admitted after deterministic train-calibrated checks.

2.2. Physics-informed generation and post-generation control

Physics-informed learning embeds governing equations, conservation laws, or domain penalties into model training [9]. In fault diagnosis, the same principle motivates spectral, envelope, order-domain, or kinematic-frequency losses for synthetic signals. Such constraints can produce high-quality samples when the generator is trainable and the physical metadata are available. However, they are less convenient for closed-source LLMs, changing prompts, or private machines whose speed/geometry metadata are incomplete.

Post-generation filtering is simpler: generate many candidates and reject implausible ones. The risk is that generic filters often check only amplitudes or low-order moments. BREEZE can be viewed as train-calibrated rejection sampling with diagnostic gates. Its novelty is not a formal physical proof; it is the integration of bearing-specific envelope evidence, train-supported spectral shape, reliability-gated current-sideband evidence, and diversity admission into an auditable black-box generation wrapper.

2.3. LLM-based signal generation and inference-time verification

LLM-based synthesis reframes time-series generation as text-conditioned reasoning or recipe generation. Its main appeal is zero-shot flexibility: a black-box LLM can generate candidate waveforms without task-specific gradient updates. Recent work has shown that language models can represent or forecast time series when numerical sequences are serialized as text [10], and domain-specific LLM signal synthesis can use recipes or code-like renderers rather than direct raw-number emission. Its weakness is that prompts are not constraints. Inference-time verifier loops, including self-refinement and rejection sampling, have become common in language tasks [11]; BREEZE adapts this idea to machinery signals, replacing subjective LLM self-critique with deterministic train-calibrated signal tests.

2.4. Bearing diagnostics as generative verification

Rolling-element faults excite structural resonances and modulate vibration at kinematically determined characteristic frequencies. Envelope analysis and spectral kurtosis are standard diagnostic tools for exposing these signatures [12, 13, 14]. Motor current signature analysis (MCSA) provides complementary evidence through sidebands around the electrical supply frequency [15]. BREEZE uses these diagnostic tools as admission gates: a generated window is not allowed into the training pool unless the same signal-processing evidence expected from real data is present in the candidate.

3. Problem Formulation

Let $\mathcal{D}_{\mathrm{tr}} = \{(x_i, y_i, m_i)\}$ denote real training windows, where $x_i \in \mathbb{R}^{C\times N}$ is a multichannel signal, y_i is the class label, and m_i contains available metadata such as speed and load. A black-box generator G proposes a candidate $\tilde{x}_r \sim G(y, m, \phi_r)$ at feedback round r, where ϕ_r is empty for the first attempt and becomes a constraint report after a rejection. The goal is to construct a train-calibrated gate family

$$\mathcal{G}_{y,m} = \{g_j(x; y, m, \mathcal{D}_{\mathrm{tr}})\}_{j=1}^{J}, \qquad g_j \in \{0, 1\}, \tag{1}$$

where every threshold used by g_j is estimated from the training split only. The gates are mixed predicates rather than a single type of one-sided inequality:

$$g_j(x) = 1 \quad \Longleftrightarrow \quad \begin{array}{ll} a_j(y,m) \le f_j(x) \le b_j(y,m) & \text{for interval features,} \\ f_j(x) \le b_j(y,m) & \text{for upper-bound tests,} \\ f_j(x) \ge a_j(y,m) & \text{for lower-bound evidence,} \\ p_{\mathrm{fault}}(x) \le b_h & \text{for healthy fault-peak suppression,} \\ d(e(x), \mathcal{S}_y) \ge \delta_y & \text{for diversity admission.} \end{array} \tag{2}$$

Here f_j denotes a deterministic signal feature, p_{fault} is an envelope-spectrum fault-frequency prominence, $e(x)$ is the physical embedding used for diversity, and $\mathcal{S}_y$ is the already admitted synthetic set for class y. A candidate is admitted if it satisfies all active gates before round K:

$$\mathcal{S}_K = \{\tilde{x} : \exists r \le K,\ \tilde{x}_r \in \mathcal{A}_{y,m}(\mathcal{S}_y)\}, \quad \mathcal{A}_{y,m} = \{x : g_j(x) = 1,\ \forall j \in \mathcal{J}_{\mathrm{active}}\}. \tag{3}$$

No learnable parameters are introduced in the verifier, and no test windows are used during calibration. Membership in $\mathcal{A}_{y,m}$ is therefore a gate-compliance statement, not a formal guarantee of physical validity or future classifier performance.

4. The BREEZE Framework

4.1. Overview

Figure 1 summarizes the closed loop. A generator emits a candidate signal or a parameter recipe that is rendered deterministically. BREEZE then executes four modules: M1 format/legality checks, M2 physical feasible-set verification, M3 diversity admission, and M4 gate-report pool logging. Failed gates are converted into a structured report containing the violated feature, observed value, train-supported range, and an executable instruction for the next generation attempt.

Figure 2 makes the responsibility boundary explicit. The LLM is responsible for proposing recipe fields, the renderer is responsible for the signal equations, and the verifier is responsible only for train-calibrated admission. This separation is important for the claims in this paper: a verifier pass is not a proof that the LLM intrinsically understands fault physics.

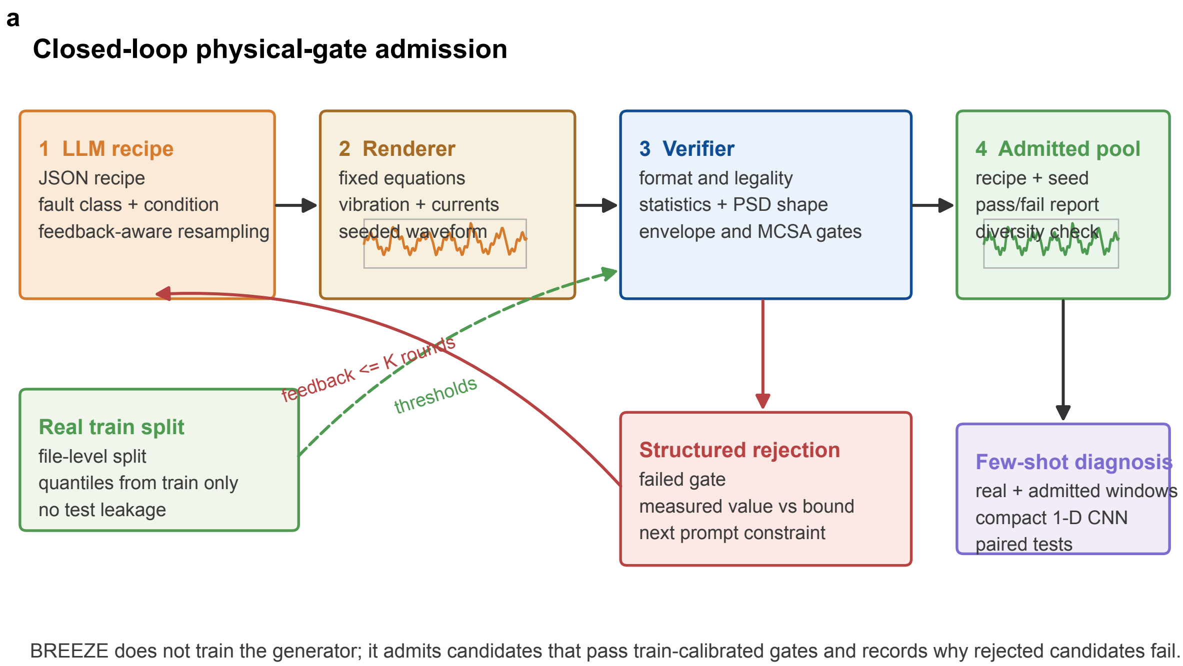

Figure 1: The BREEZE closed-loop physical-gate admission framework. The LLM proposes a recipe, the renderer deterministically constructs a candidate window, and the verifier admits or rejects it using gates calibrated only from the real training split. Rejected candidates return structured feedback; passed candidates enter a gate-admitted pool with an auditable report.

4.2. M1: format and legality

M1 rejects malformed outputs: wrong shape, non-finite values, near-constant channels, and long repeated segments. In the PU study the expected shape is $(3, 2048)$ after selecting one vibration channel and two phase-current channels. In the private machine-tool study the schema is $(4, 2048)$ for three vibration axes plus current.

4.3. M2: physical feasible-set gates

Boundary-restricted statistics. For each class and channel, BREEZE estimates train-split quantile intervals for RMS, peak value, standard deviation, kurtosis, skewness, and crest factor. The original verifier uses per-feature intervals. The v2 verifier uses a union domain: a candidate passes if it lies inside either the legacy quantile box or a robust axis ellipsoid defined by median/IQR-standardized feature distances. This prevents a physically plausible candidate from being rejected solely because one correlated statistic is slightly outside a marginal boundary.

Spectral shape and PSD distance. The original verifier uses hard Welch-PSD band fractions. v2 replaces this with overlapping triangular bands and a normalized PSD-CDF Wasserstein-1 distance to the class train reference. These features are still deterministic and train-calibrated, but they avoid discontinuities when a spectral peak lies near a hard band boundary.

Envelope-spectrum evidence. For fault classes, the vibration signal is band-passed through train-supported resonance bands, demodulated using the Hilbert transform, and checked for a prominent peak near the class characteristic frequency. Healthy candidates must not display a fault-frequency envelope peak above the healthy threshold. v2 uses the top three train-supported resonance bands instead of a single spectral-kurtosis maximum.

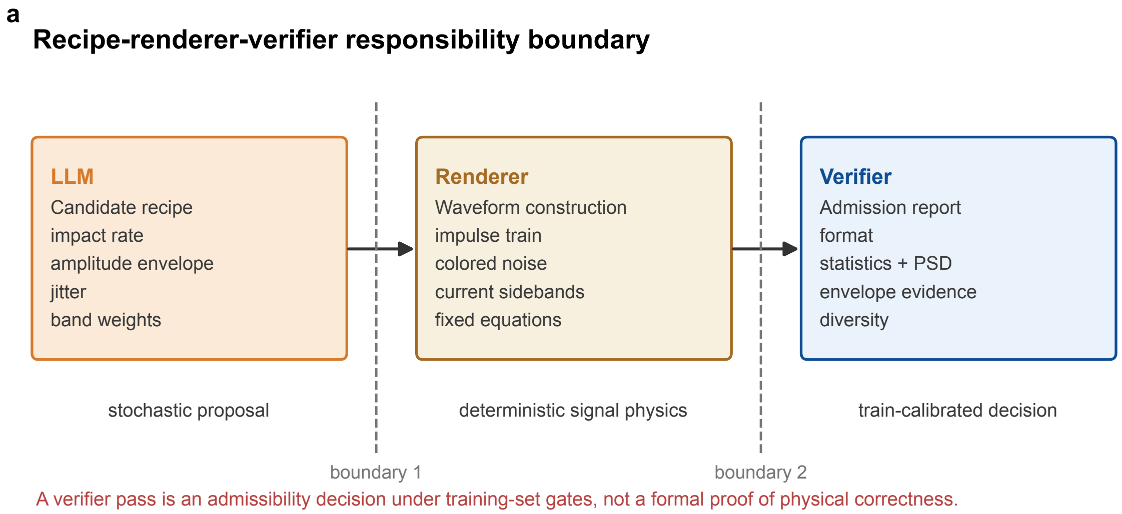

Figure 2: Recipe-renderer-verifier responsibility boundary. The stochastic LLM proposal, deterministic signal construction, and train-calibrated gate decision are separate components. The verifier can reject and report constraint violations, but it does not repair a waveform or provide a formal physical correctness certificate.

Current-sideband evidence. For the PU setting, two phase-current channels are combined into a vector-current MCSA score. If the train split shows that sidebands are not separable from healthy current spectra, the MCSA score is recorded in the gate report but not used as a hard gate. This avoids enforcing a constraint that the real train data do not support.

4.4. M3: diversity admission

The admitted pool should not be a set of near-duplicates. BREEZE computes a physical embedding from statistics, spectral fractions, envelope scores, and current-sideband features. Candidate windows whose nearest-neighbour distance to the accepted pool is below a class-wise real-real reference threshold are not admitted. The threshold is derived from the train split.

4.5. M4: gate-report resampling

Algorithm 1 defines the closed loop. The verifier never repairs a failed waveform. It either admits the generated candidate as-is with a gate-compliance report, or rejects it and returns a report to guide another generator call.

Algorithm 1 BREEZE closed-loop physical-gate admission

Require: class y, metadata m, generator G, verifier $\mathcal{V}$, maximum feedback rounds K

$\phi_0 \leftarrow \varnothing$
for $r = 0$ to K **do**
  $\tilde{x}_r \sim G(y, m, \phi_r)$
  $(p_r, R_r) \leftarrow \mathcal{V}(\tilde{x}_r, y, m)$
  if $p_r = 1$ **then**
    return admitted sample and gate report R_r
  end if
  $\phi_{r+1} \leftarrow \mathrm{ReportToFeedback}(R_r)$
end for
return rejection

Table 1
PU file-split window counts under N09_M07_F10.

Split	healthy	OR	IR	total
Train	1200	1202	1444	3846
Test	300	301	361	962

5. Experimental Setup

5.1. Paderborn benchmark

The main benchmark is the Paderborn University bearing dataset [16] under condition N09_M07_F10 (900 rpm, 0.7 Nm torque, 1000 N radial load). Raw 64 kHz signals are decimated to 8 kHz using zero-phase FIR filtering. Windows have 2048 samples. The retained channels are 'vibration_1', 'phase_current_1', and 'phase_current_2'; internal short names 'X/Y/Z' refer to these retained channels and should not be interpreted as three-axis PU vibration. The file-based split uses the first 80% of measurement files for training and the last 20% for testing for every bearing. It contains 3846 train windows and 962 test windows, with no file-ID overlap.

5.2. Private machine-tool dataset

A private laboratory machine-tool dataset is used to test schema-level verifier portability. Internal documentation describes a CNC machining acquisition system driven by a standardized G-code process. The signals are collected by an NI acquisition card at 4 kHz with four synchronous analog channels: $X/Y/Z$ vibration and spindle load Current. The documented dataset covers normal machining, lead-screw anomaly, and base imbalance. In the present BREEZE workspace, however, local file names identify three classes only as 1, 2, and 3, and the available documentation does not unambiguously map these prefixes to the physical state names. We therefore report them as MT-1, MT-2, and MT-3. The BREEZE windows have 2048 samples with stride 1024, corresponding to 0.512 s windows and 0.256 s hops at 4 kHz. Train files are 1, 2, 4, 5, and 10; test files are 7 and 8. Because machine geometry, operating speed, and the prefix-to-state mapping are unavailable in the audited workspace metadata, the machine-tool verifier uses only generic train-calibrated statistics and normalized spectral constraints.

5.3. Baselines and classifier

The PU downstream study compares real-only training, noise augmentation, VAE and GAN baselines, few-shot VAE/GAN baselines, open-loop basic LLM generation, open-loop physics-guided generation, stats-only filtering, envelope-only filtering, the original BREEZE pools for $K = 0, 1, 2, 3$, and the v2 offline-rescreened pool. The classifier is a compact 1D CNN trained for 60 epochs. Few-shot settings use 10, 25, or 50 real windows per class and eight random seeds. Synthetic pools are capped at 200 windows per class when added to a few-shot training set. Accuracy, macro-F1, per-class F1 where available, physical MMD, nearest-neighbour diversity, acceptance rate, and paired Wilcoxon tests are reported. The v2 pool is not a new v2 LLM generation run; it is a stricter offline rescreening of the frozen original K=3 generated candidates.

6. Results

6.1. Closed-loop acceptance

The original physics-guided closed loop generated 450 slots. Here a slot means one requested LLM generation unit for a class. A feasible slot may contribute the selected round window and zero or more feasible expansion windows produced from the same slot recipe. As K increases, the accepted-slot rate rises while the average number of LLM calls remains near two at $K = 3$ (Table 2). The v2 offline rescreening is stricter: it admits 286 of 450 slots, produces 761 accepted items before diversity, keeps 757 generated windows after diversity, and yields class counts 192/304/261 for healthy/OR/IR. Since v2 does not call the LLM again, the LLM-call cost in Table 2 belongs only to the original closed-loop run.

Table 3 and Figure 3 summarize why v2 rejects terminal slots. Counts can exceed the number of rejected slots because one terminal candidate may violate multiple gates. Healthy rejections are dominated by forbidden fault-frequency envelope peaks, while OR/IR rejections are dominated by the statistical union domain and soft spectral

Table 2
Closed-loop acceptance and cost for the original physics-guided run.

K	accepted slots	acceptance	mean LLM calls/slot
0	253/450	0.5622	1.0000
1	295/450	0.6556	1.4378
2	330/450	0.7333	1.7822
3	355/450	0.7889	2.0489

Table 3
Terminal rejected-slot failure counts in v2 offline rescreening.

Class	rejected slots	stats	soft spectrum	envelope / PSD-W1
healthy	60	4	0	51 / 10
OR	60	45	21	0 / 0
IR	44	43	7	0 / 0

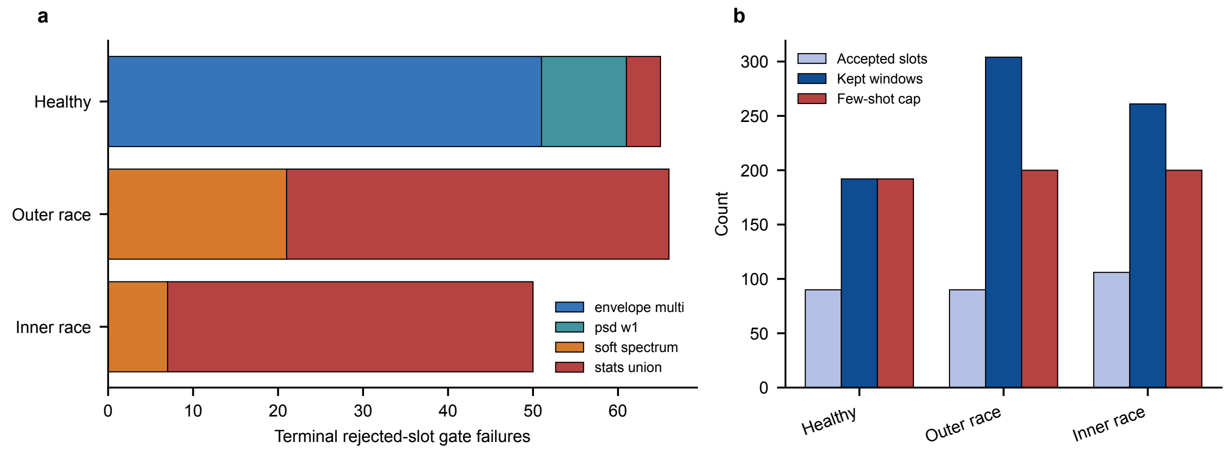

Figure 3: Failure and slot-window accounting for v2 offline rescreening. Panel a shows terminal rejected-slot gate failures by class. Panel b shows how accepted slots expand into kept windows and the 200-window/class few-shot cap. Source data are the v2 fail-reason and slot-window mapping CSV files.

shape. This failure pattern is useful for diagnosing which parts of the LLM recipe family remain weak; it should not be interpreted as a guarantee that all passed samples are physically correct.

Figure 4 gives a concrete rejected candidate from the same audit trail. The slot is labelled healthy by the generator output, but its envelope spectrum contains forbidden IR-frequency evidence above the train-calibrated healthy limit. This is a single diagnostic example, not a distributional result; the aggregate failure counts are reported in Table 3.

6.2. Coverage sensitivity

Table 4 reports train/test pass rates for v1 verifier coverages c85, c90, and c95. Pass rates change modestly because the hard spectrum gate is unchanged across these calibrations; this motivated the v2 smooth-spectrum and PSD-W1 redesign.

6.3. Physical fidelity and diversity

Figure 5 compares real and gate-admitted synthetic examples in time, FFT, and envelope-spectrum domains. Figure 6 shows the effect of admission on RMS and kurtosis distributions. Figure 7 and Table 5 summarize feature-MMD and nearest-neighbour diversity. MMD2 is computed on an eight-dimensional vibration-channel feature vector:

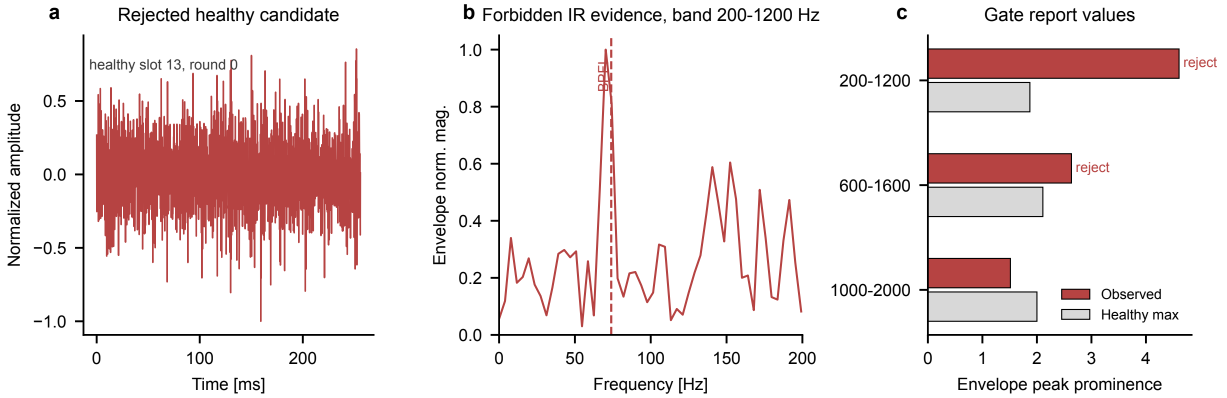

Figure 4: Example of a rejected healthy candidate. Panel a shows the candidate waveform. Panel b shows the envelope spectrum in the resonance band that triggered forbidden IR-frequency evidence. Panel c reports the observed envelope peak prominence against the train-calibrated healthy upper limit for the three IR resonance bands. The source record is `runs/rescreen_v2_full/records/healthy_0013.json`.

Table 4
PU verifier coverage sensitivity. Rates are recomputed on frozen train/test splits; thresholds are calibrated without using test data.

Coverage	split	all	healthy	OR	IR	spectrum fails
c85	train	0.9285	0.9300	0.9285	0.9273	104
c85	test	0.8992	0.9100	0.8970	0.8920	44
c90	train	0.9368	0.9425	0.9326	0.9356	104
c90	test	0.9148	0.9400	0.9103	0.8975	44
c95	train	0.9441	0.9467	0.9443	0.9418	104
c95	test	0.9210	0.9400	0.9169	0.9086	44

RMS, kurtosis, crest factor, skewness, and Welch-PSD band-energy fractions over 0–500, 500–1500, 1500–2500, and 2500–4000 Hz. For each class, up to 150 synthetic windows and 150 train-real windows are standardized by the sampled real mean and standard deviation, and RBF-kernel MMD2 is computed with the median pairwise-distance bandwidth heuristic. v2 improves MMD over the original BREEZE $K = 3$ pool for OR and IR, but not for healthy. Open-loop physics-guided and envelope-only pools remain competitive on this particular feature-MMD metric, so physical-gate admission should be interpreted jointly with gate reports and downstream utility rather than as a universal MMD minimizer.

6.4. Downstream few-shot diagnosis

Figure 8 and Table 6 report downstream accuracy. The v2 offline-rescreened pool outperforms real-only training in all three few-shot regimes, and these three comparisons remain significant after Benjamini–Hochberg correction across the full family of 42 v2-vs-main-baseline paired tests. However, v2 is not uniformly higher than all conventional baselines: envelope-only is higher at 10-shot, open-loop physics-guided and noise augmentation are higher at 25-shot, and noise augmentation is strongest at 50-shot in the frozen main table. This is an important boundary of the current evidence.

6.5. Feedback-round ablation

Figure 9 shows the acceptance-cost trade-off and the downstream accuracy of the original BREEZE $K = 0, \ldots, 3$ pools. Increasing K raises admission probability, but downstream accuracy is not monotonic. This supports the paper's central distinction: physical admission and diagnostic utility must be evaluated separately.

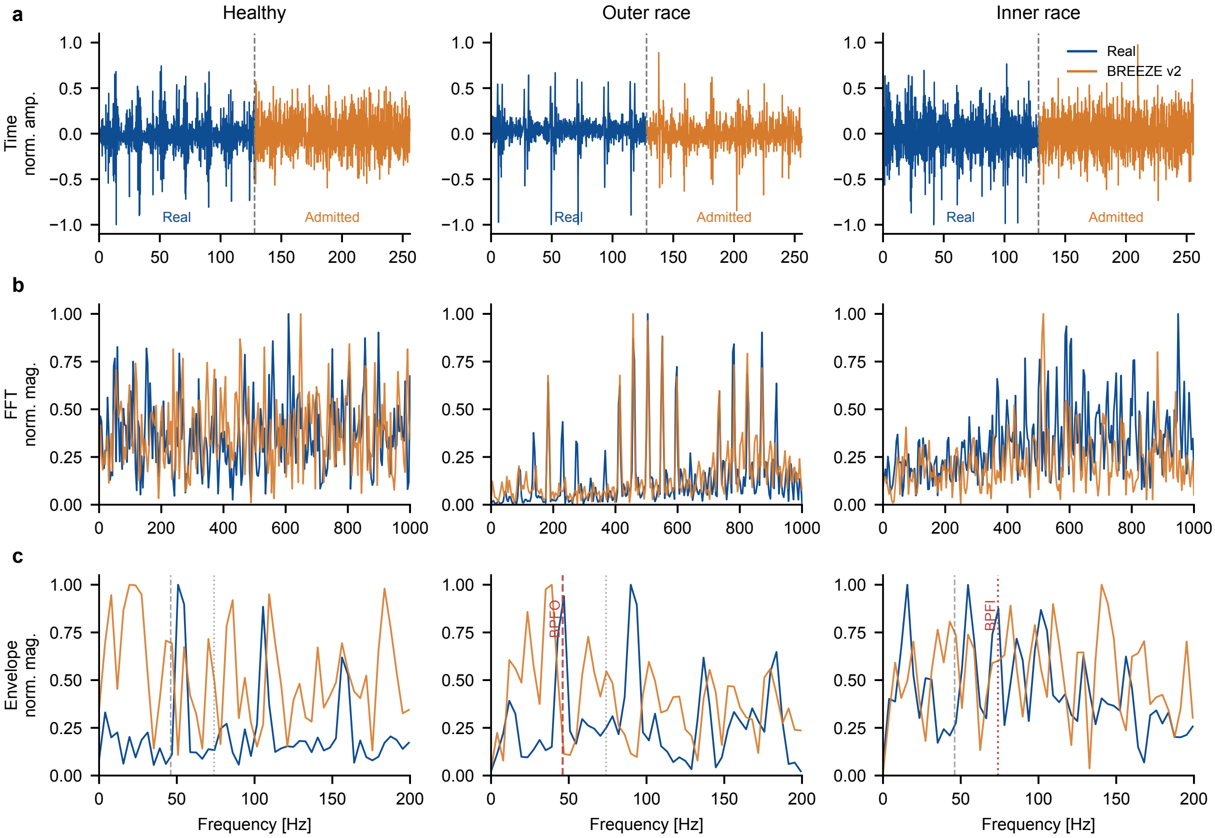

Figure 5: Representative real and v2 gate-admitted synthetic windows in the time, FFT, and envelope-spectrum domains. Blue traces denote real train windows and orange traces denote admitted synthetic windows. Vertical reference lines mark BPFO and BPFI; the class-relevant fault frequency is highlighted in red.

6.6. PU cross-condition verifier audit

To address operating-condition dependence without inventing an unrun cross-condition LLM experiment, we also instantiate the v2 verifier separately on all four preprocessed PU conditions and evaluate real train/test pass rates. Figure 10 and Table 8 report compact class-group summaries. This is a verifier coverage audit only: no synthetic pool is generated for these extra conditions. The results show that fault-class windows generally remain within the train-calibrated gates, while the healthy class is conservatively rejected more often because the forbidden fault-frequency envelope gate is strict.

6.7. Private machine-tool schema audit

Table 9 summarizes the private dataset. The four-channel verifier uses normalized PSD features rather than PU bearing kinematics. The sampling rate is known to be 4 kHz, but absolute machine-order gates are not used because the workspace lacks the corresponding operating-speed and geometry metadata. Train pass rates are 0.938–0.959. Test pass rates reveal distribution shift in MT-1 and MT-3, especially test file 1_7; this is useful diagnostic information rather than a result to hide. Real-only downstream accuracy rises from 0.3431 ± 0.0321 at 10 real windows per class to 0.7006 ± 0.0626 at 50 real windows per class. No machine-tool synthetic pool has been generated and screened yet, so no augmentation gain is claimed on this dataset.

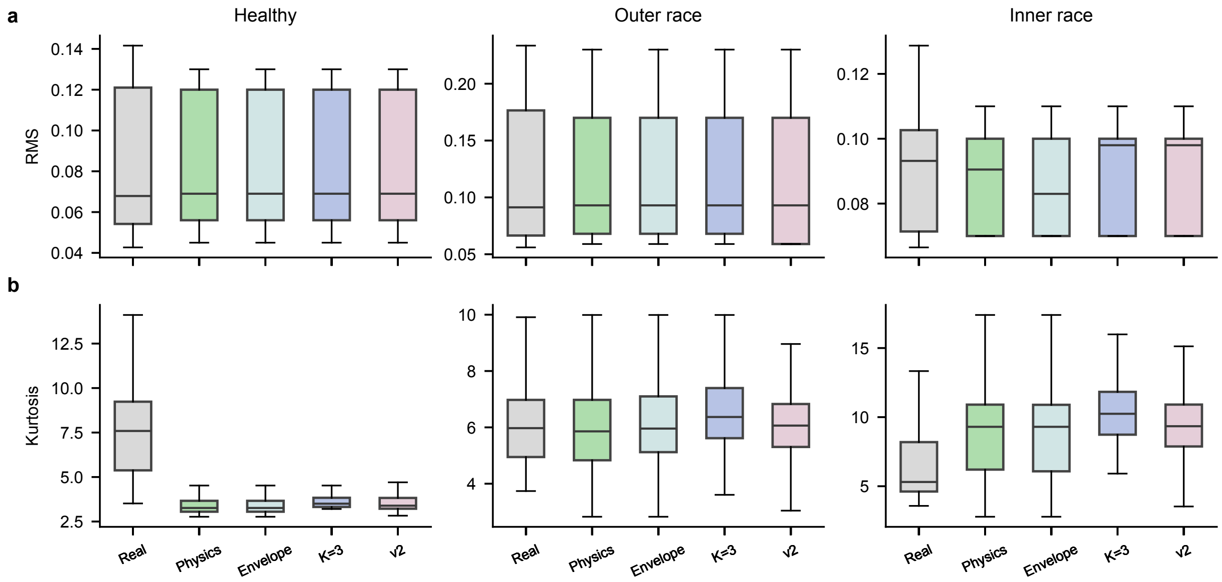

Figure 6: RMS and kurtosis distributions for real windows and synthetic pools. The v2 pool is included when available.

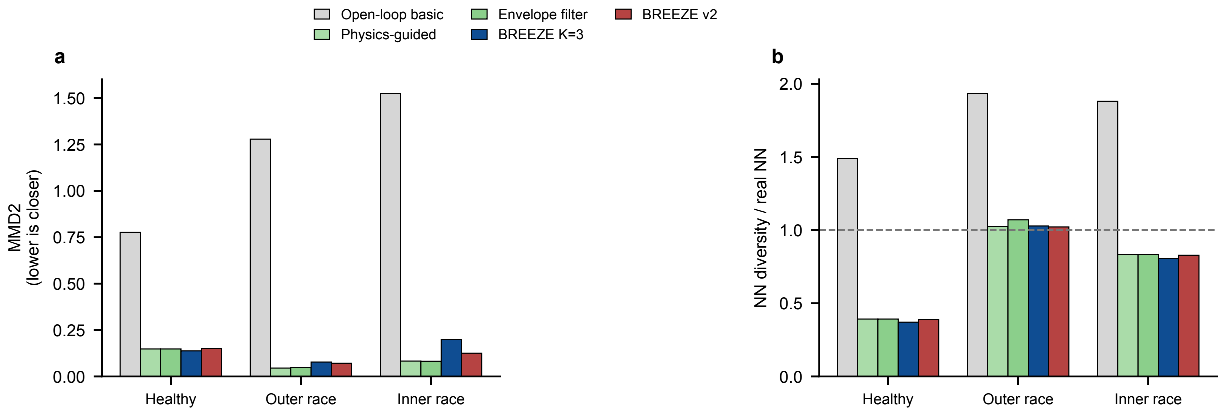

Figure 7: Feature-distance and diversity summaries for synthetic pools. Panel a reports MMD2 on standardized eight-dimensional vibration features. Panel b reports nearest-neighbour diversity normalized by the corresponding real-real reference. Lower MMD2 is closer to the sampled real feature distribution; a diversity ratio near one indicates real-like spacing under this feature set.

7. Discussion

Why physical-gate admission rather than generator training? BREEZE is useful when the generator is expensive, closed source, or expected to change. The verifier is deterministic and calibrated from the train split; therefore, the same admission logic can wrap an LLM recipe generator today and a different black-box generator later.

Physical fidelity is not identical to downstream accuracy. The experiments show that gate-admitted pools can improve real-only diagnosis and the original BREEZE baseline in low-data settings, but strong conventional

Table 5
Physical fidelity and diversity. Lower MMD2 is closer to real features; NN diversity should be interpreted relative to the real-real reference.

Pool	class	n	MMD2	NN div.	real NN div.
open-loop basic	healthy	135	0.7769	1.3263	0.8910
open-loop basic	OR	142	1.2786	0.7972	0.4122
open-loop basic	IR	139	1.5248	1.6169	0.8598
open-loop phys.	healthy	150	0.1481	0.3495	0.8910
open-loop phys.	OR	150	0.0451	0.4224	0.4122
open-loop phys.	IR	150	0.0829	0.7162	0.8598
BREEZE $K = 3$	healthy	519	0.1378	0.3305	0.8910
BREEZE $K = 3$	OR	638	0.0776	0.4239	0.4122
BREEZE $K = 3$	IR	390	0.1990	0.6919	0.8598
BREEZE v2	healthy	192	0.1507	0.3468	0.8910
BREEZE v2	OR	304	0.0713	0.4212	0.4122
BREEZE v2	IR	261	0.1254	0.7123	0.8598

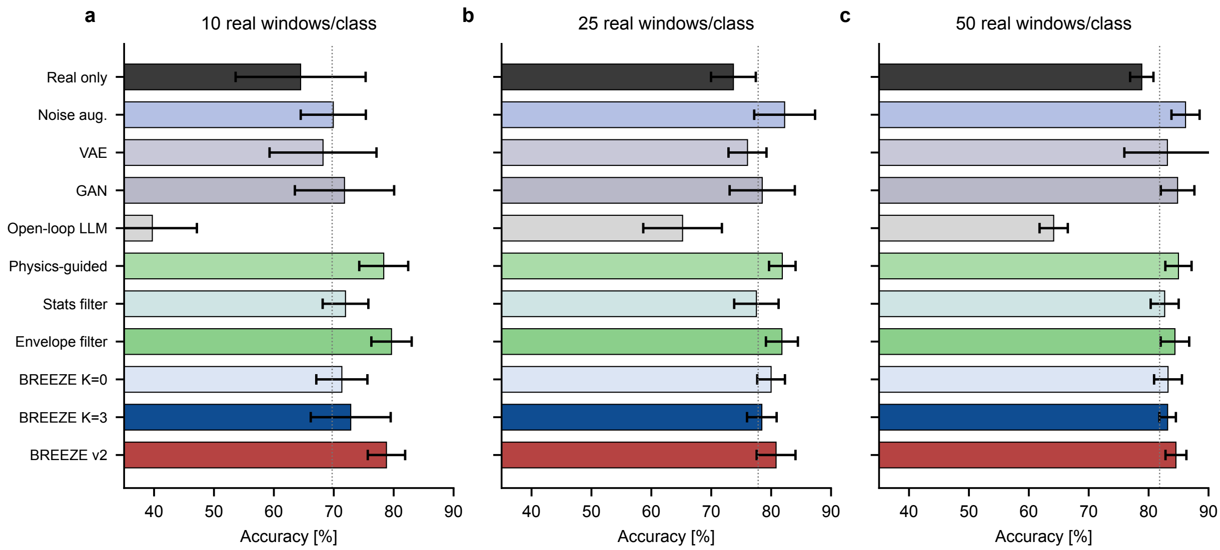

Figure 8: Downstream test accuracy for selected baselines. The final bar shows the v2 offline-rescreened pool when the v2 evaluation file is present.

augmentation can still win in some higher-shot regimes. This does not invalidate physical-gate admission; it clarifies the contribution. BREEZE provides auditable control over which generated samples are admitted, not a guarantee that physical filtering alone will dominate every classifier-training baseline.

Threshold calibration. All thresholds are train-only. The c85/c90/c95 sensitivity study shows modest changes in real pass rates and exposes a hard-spectrum limitation in the v1 verifier. The v2 redesign addresses this through smooth spectral features and PSD-W1. The additional four-condition PU audit shows that v2 can be calibrated under multiple operating conditions, but it also exposes a conservative healthy fault-peak suppression gate. Therefore, a full cross-condition augmentation claim still requires new generated pools and downstream evaluation.

Circular validation risk. The verifier and the deterministic renderer both use bearing-domain concepts. This raises a possible circular-validation risk: a generator may learn to satisfy the measured gates without matching all aspects of the real fault distribution. We reduce this risk by reporting downstream diagnosis, MMD, diversity, pass/fail records,

Table 6
Selected PU downstream results. Values are mean ± standard deviation over eight seeds. The v2 pool uses the 200-window/class synthetic cap, giving 192+200+200=592 synthetic windows in each few-shot run.

Baseline	Metric	10/class	25/class	50/class
real only	Acc.	0.6447 ± 0.1085	0.7370 ± 0.0374	0.7886 ± 0.0195
open-loop phys.	Acc.	0.7834 ± 0.0408	0.8186 ± 0.0221	0.8498 ± 0.0219
envelope-only	Acc.	0.7964 ± 0.0337	0.8180 ± 0.0267	0.8439 ± 0.0238
noise aug.	Acc.	0.6992 ± 0.0544	0.8225 ± 0.0508	0.8616 ± 0.0235
BREEZE $K = 3$	Acc.	0.7283 ± 0.0667	0.7844 ± 0.0247	0.8317 ± 0.0138
BREEZE v2	Acc.	0.7880 ± 0.0312	0.8082 ± 0.0324	0.8456 ± 0.0174
real only	Macro-F1	0.6040 ± 0.1292	0.7184 ± 0.0425	0.7743 ± 0.0253
BREEZE v2	Macro-F1	0.7906 ± 0.0289	0.8079 ± 0.0354	0.8484 ± 0.0178

Table 7
Selected paired Wilcoxon tests on accuracy for BREEZE v2. The q column is Benjamini–Hochberg adjusted across the full family of 42 v2-vs-main-baseline tests; the complete table is provided in the revision statistics output.

Comparison	n/class	Delta	p	q
v2 vs real only	10	+0.1433	0.0078	0.0328
v2 vs real only	25	+0.0712	0.0156	0.0438
v2 vs real only	50	+0.0570	0.0078	0.0328
v2 vs BREEZE $K = 3$	10	+0.0597	0.0156	0.0438
v2 vs BREEZE $K = 3$	25	+0.0238	0.0547	0.1209
v2 vs BREEZE $K = 3$	50	+0.0139	0.0391	0.0965
v2 vs envelope-only	10	-0.0084	0.2109	0.3281
v2 vs open-loop phys.	25	-0.0104	0.7422	0.8203
v2 vs noise aug.	50	-0.0160	0.1484	0.2494

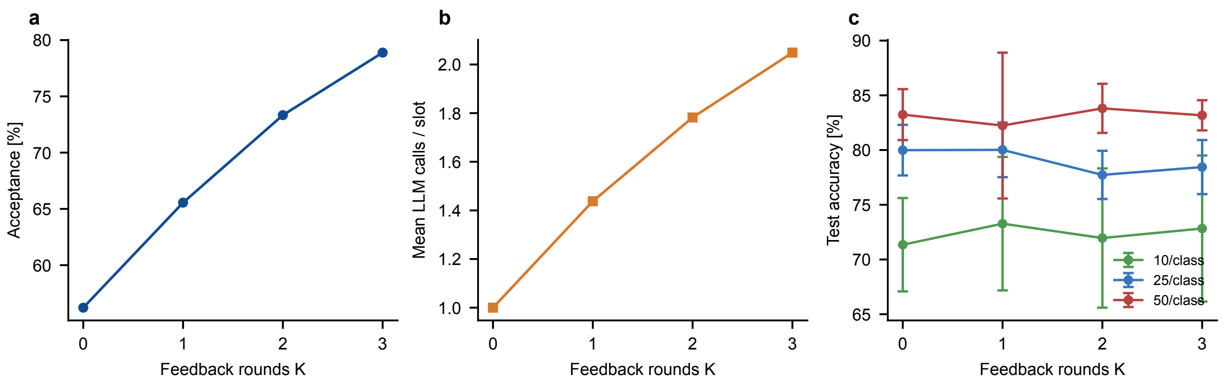

Figure 9: Original closed-loop acceptance, average LLM calls, and downstream accuracy across feedback rounds K.

and strong augmentation baselines, but external physical validation and human diagnostic review remain necessary before industrial deployment.

Limitations. The current PU results focus on one operating condition. The private machine-tool study verifies schema-level portability, but not augmentation, because no audited private synthetic pool exists yet. The class semantics of the private data must be provided before physical fault names can be written. The current downstream study uses a

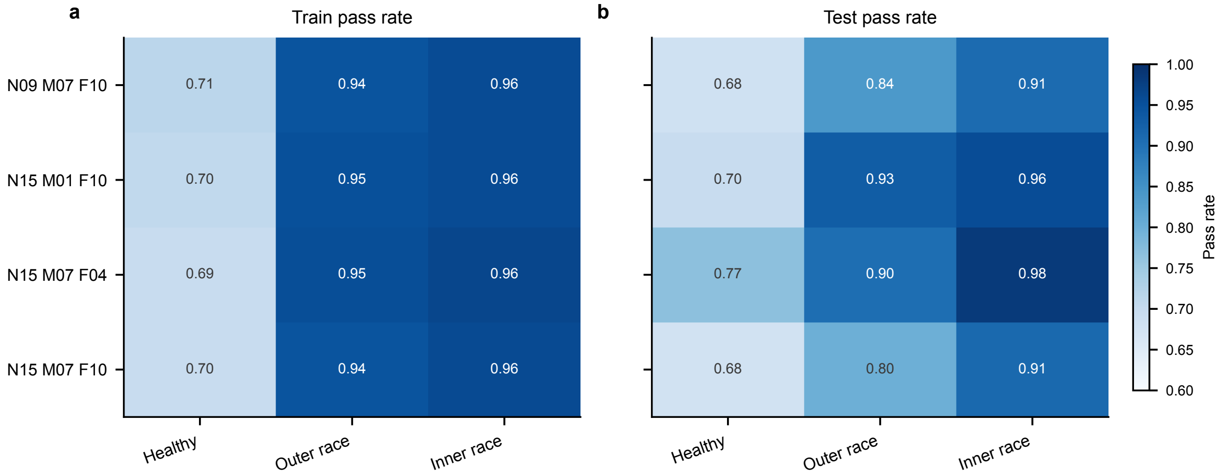

Figure 10: PU cross-condition v2 verifier audit. Heatmaps show train and test pass rates after recalibrating the verifier within each condition. This figure audits gate coverage on real windows only and does not report cross-condition synthetic augmentation.

Table 8
PU cross-condition v2 verifier audit. Fault mean is the average of OR and IR pass rates. The audit is not a cross-condition augmentation experiment.

Condition	train healthy	train fault mean	test healthy	test fault mean
N09_M07_F10	0.7125	0.9507	0.6800	0.8729
N15_M01_F10	0.7049	0.9533	0.6967	0.9428
N15_M07_F04	0.6938	0.9595	0.7667	0.9433
N15_M07_F10	0.6958	0.9516	0.6777	0.8545

Table 9
Private machine-tool results. The documented dataset contains normal machining, lead-screw anomaly, and base imbalance, but labels are reported anonymously because the local files do not map prefixes 1/2/3 to those physical state names.

Block	class/setting	n	pass or Acc.	macro-F1
Verifier train	MT-1	675	0.9585	–
Verifier train	MT-2	383	0.9530	–
Verifier train	MT-3	679	0.9381	–
Verifier test	MT-1	166	0.7590	–
Verifier test	MT-2	166	0.9337	–
Verifier test	MT-3	154	0.7727	–
Real-only 10/class	all	8 seeds	0.3431 ± 0.0321	0.3368 ± 0.0318
Real-only 25/class	all	8 seeds	0.4964 ± 0.0566	0.4897 ± 0.0530
Real-only 50/class	all	8 seeds	0.7006 ± 0.0626	0.7021 ± 0.0632

compact 1D CNN; additional classifiers are needed to test model-level robustness. Finally, exact citation metadata for unpublished or in-house MechaForge assets must be completed before submission.

8. Conclusion

This paper introduced BREEZE, a training-free closed-loop framework for physical-gate admission of LLM-generated fault-diagnosis signals. Instead of training a new generator or repairing failed waveforms, BREEZE tests each candidate against train-calibrated signal-processing gates, feeds structured failure reports back for resampling, and admits only samples with auditable gate reports. On the Paderborn benchmark, the v2 offline-rescreened pool improves few-shot real-only diagnosis after multiple-test correction, while physical metrics and strong baselines reveal where the method is competitive rather than dominant. Additional PU operating-condition audits and the private machine-tool study show how the verifier can be reconfigured, but not yet that synthetic augmentation transfers across all machines and conditions. Future work will add full cross-speed/load generation loops, audited synthetic pools for the private dataset, additional diagnostic classifiers, and broader comparisons with trainable physics-informed generators.

Reproducibility Notes

The numerical results in this manuscript are traceable to the following local outputs:

- `results/downstream_file.csv`
- `results/custom_pool_eval.csv`
- `results/pool_metrics.csv` and `results/pool_metrics_v2.csv`
- `results/calibration_sensitivity.csv`
- `results/revision_v2_significance_all_baselines_bh.csv`
- `results/revision_v2_slot_window_mapping.csv`
- `results/revision_v2_rejected_slot_fail_reasons.csv`
- `results/cross_condition_verifier_full.csv`
- `results/mt_verifier_real_pass.csv`
- `results/mt_real_only_eval.csv`

Snapshot reports are stored in the project `reports/` directory.

Data Availability Statement

The Paderborn University bearing dataset is publicly available from the data provider. The private machine-tool dataset contains laboratory data and is not publicly released in the current submission; derived aggregate metrics, verifier summaries, and split descriptions are provided in the supplementary material. Code, prompts, gate reports, and result tables can be shared subject to institutional and data-use constraints.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work, the authors used OpenAI Codex to assist with code inspection, experiment scripting, result-table assembly, figure generation, and manuscript drafting. After using this tool, the authors reviewed, verified, and edited the content and take full responsibility for the final manuscript.

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